

Building a cross-border marine monitoring network to assess the impacts of climate change in Ireland's seas

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A climate change impact monitoring and assessment toolbox for Irish Seas

The two jurisdictions of Ireland's marine area (RoI and Northern Ireland/UK), require coordinated monitoring.

Management of shared marine resources is limited by gaps in spatial data on key pelagic ecosystem components (phytoplankton, zooplankton, seabirds, cetaceans).

Improved biological and environmental data are essential for developing spatially integrated ecosystem models to support fisheries and marine management.

ACCLIMATISE, funded by the SEUPB PEACEPLUS, is building a cross-border marine monitoring toolbox combining ship-based, remote, and autonomous sampling to inform a spatially integrated ecosystem model for Irish Seas. Based on these outputs ACCLIMATISE will develop future decision-support tools for marine planning in Ireland's seas.

Assessment of plankton-climate interactions in Irish seas

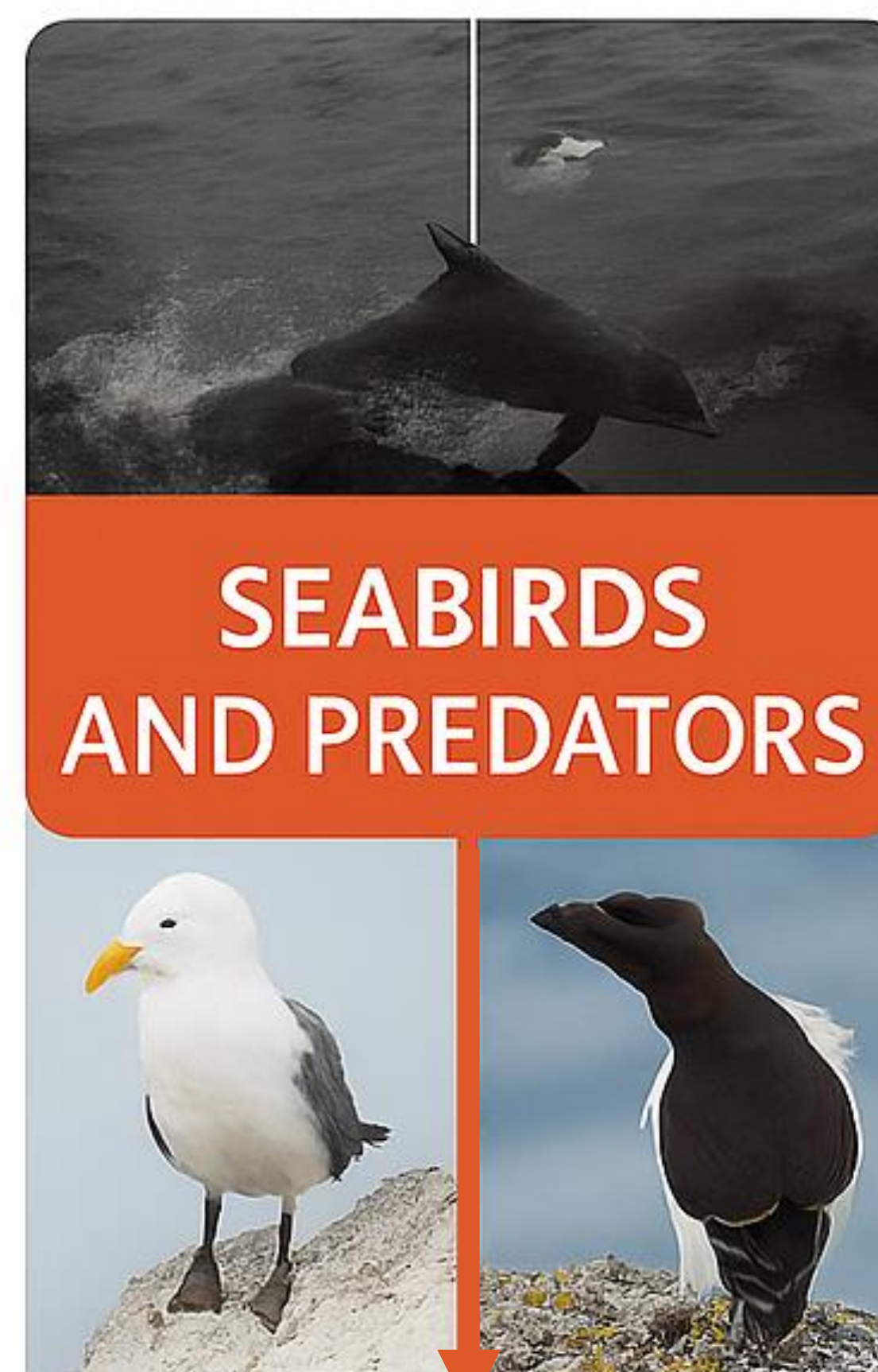
- Develop an all-island plankton network using standardized protocols.
- Pelagic habitat mapping, combining high-frequency oceanographic and plankton sampling.
- Test novel technologies for high-frequency plankton surveys : including high resolution imaging systems and metagenomic approaches.
- Deployment of oceanographic instruments on fishing vessels across both jurisdictions.

Spatially dynamic ecosystem model for Irish Seas

- Spatial ecosystem model for the Western Irish Sea to support cross-border marine planning
- Integrated datasets combining multi-trophic data with long-term AFBI, MI, ICES & Copernicus records
- Food-web & ecosystem processes resolved across space and time for climate, biodiversity, energy & fisheries pressures
- Co-develop modelling scenarios with fisheries stakeholders across both jurisdictions

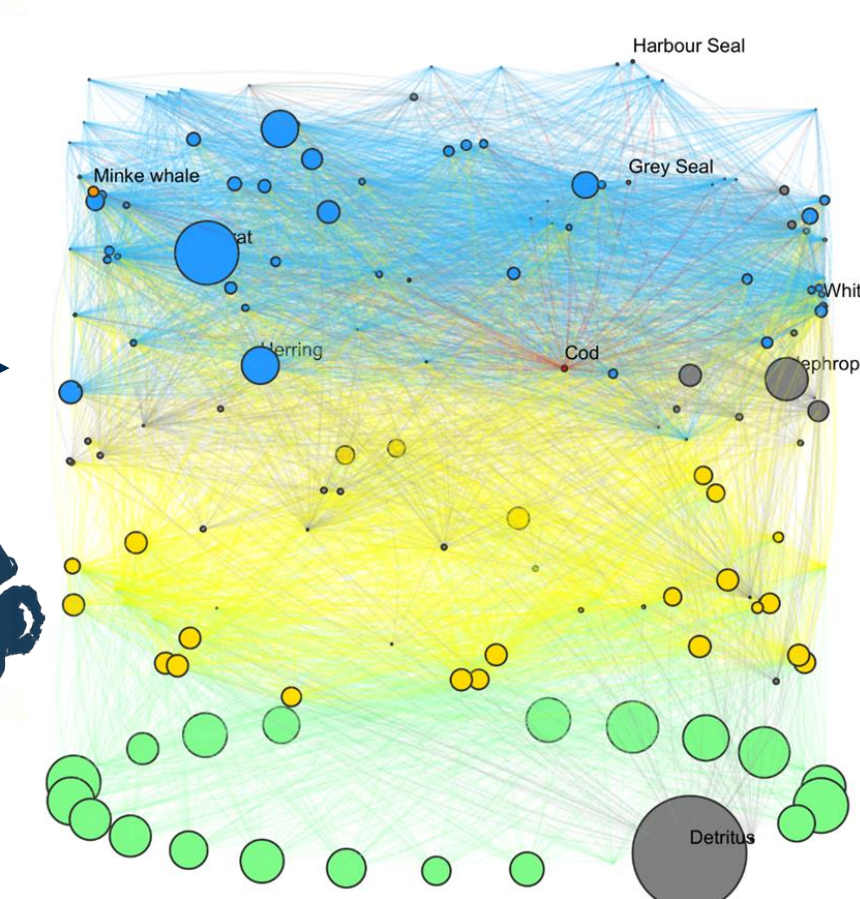
DATA CURATION

PLANKTON-CLIMATE INTERACTIONS



SPATIALLY DYNAMIC ECOSYSTEM MODEL FOR THE IRISH SEA

EXISTING FISHERIES AND ENVIRONMENTAL MONITORING



Developing transboundary conservation strategies for seabirds and marine mammals

- Use telemetry to map seabird distributions, foraging areas and priority-species needs.
- Use citizen-science and cross-border data to assess bottlenose dolphin connectivity and abundance.
- Trial passive acoustic monitoring to improve knowledge of scarce nocturnal breeding seabird and marine mammal distributions.
- Develop open-source tools for cross-border monitoring of vocalizing species and guiding resource allocation.

Data Curation & QA

- Coordinate data governance and data stewardship across AFBI, Marine Institute and all project partners
- Ensure efficient data flow between work packages and transparent access to support project outputs
- Publish metadata to international standards and ensure datasets meet FAIR principles
- Assign DOIs and transfer datasets to national data centres (e.g. MEDIN, Marine Institute, BODC).

Key Outputs

- All-island plankton monitoring strategy using aligned methods, high-frequency sampling and new technologies (imaging-based and metagenomics).
- Citizen Science Fisheries Research Fleet and integrated datasets to map climate-driven plankton dynamics.
- Cross-border action plans for seabirds and bottlenose dolphins supported by new tracking, PAM and metagenomic data.
- Risk maps and transboundary monitoring for seabirds and cetaceans, including platforms-of-opportunity and photo-ID catalogues.
- Spatially dynamic ecosystem model integrating data from plankton to predators across the Western Irish Sea.
- FAIR-compliant data governance with ISO-standard metadata and an open project data catalogue.
- Long-term data legacy via DOIs, standardization and transfer to national marine data centers (e.g. MEDIN, Marine Institute, BODC).